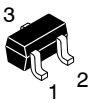
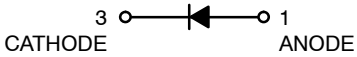
# Switching Diode

# BAS116TT1G




**CASE 463**
**SOT−416**
**STYLE 2**




### Features

- Low Leakage Current Applications
- Medium Speed Switching Times
- Available in 8 mm Tape and Reel
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant

### MAXIMUM RATINGS ($T_A$ = 25°C)

| Rating | Symbol | Max | Unit |
|---|---|---|---|
| Continuous Reverse Voltage | $V_R$ | 75 | V |
| Peak Forward Current | $I_F$ | 200 | mA |
| Peak Forward Surge Current | $I_{FM(surge)}$ | 500 | mA |

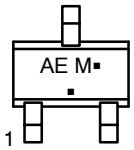
### THERMAL CHARACTERISTICS

| Characteristic | Symbol | Max | Unit |
|---|---|---|---|
| Total Device Dissipation, FR−4 Board (Note 1) $T_A$ = 25°C<br>Derated above 25°C | $P_D$ | 225<br>1.8 | mW<br>mW/°C |
| Thermal Resistance, Junction−to−Ambient (Note 1) | $R_{\theta JA}$ | 555 | °C/W |
| Total Device Dissipation, FR−4 Board (Note 2) $T_A$ = 25°C<br>Derated above 25°C | $P_D$ | 360<br>2.9 | mW<br>mW/°C |
| Thermal Resistance, Junction−to−Ambient (Note 2) | $R_{\theta JA}$ | 345 | °C/W |
| Junction and Storage Temperature Range | $T_J$, $T_{stg}$ | −55 to +150 | °C |

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

1. FR−4 @ Minimum Pad
2. FR−4 @ 1.0 × 1.0 Inch Pad

### MARKING DIAGRAM




AE = Specific Device Code
M = Date Code
▪ = Pb−Free Package

### ORDERING INFORMATION

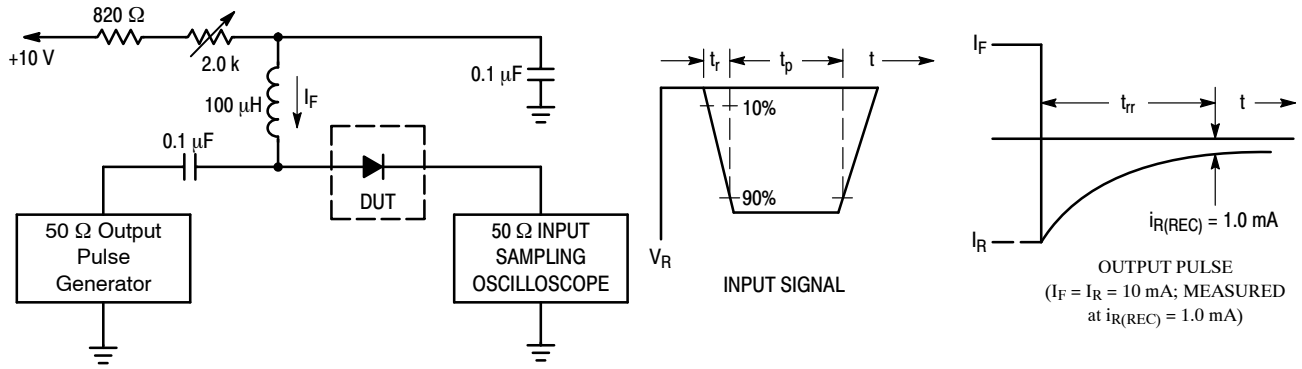
| Device | Package | Shipping† |
|---|---|---|
| BAS116TT1G | SOT−416 (Pb−Free) | 3000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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**ELECTRICAL CHARACTERISTICS** ($T_A$ = 25°C unless otherwise noted)

| Characteristic | Symbol | Min | Max | Unit |
|---|---|---|---|---|
| **OFF CHARACTERISTICS** | | | | |
| Reverse Breakdown Voltage ($I_{BR}$ = 100 μAdc) | $V_{(BR)}$ | 75 | – | Vdc |
| Reverse Voltage Leakage Current ($V_R$ = 75 Vdc)<br>($V_R$ = 75 Vdc, $T_J$ = 150°C) | $I_R$ | –<br>– | 5.0<br>80 | nAdc |
| Forward Voltage ($I_F$ = 1.0 mAdc)<br>($I_F$ = 10 mAdc)<br>($I_F$ = 50 mAdc)<br>($I_F$ = 150 mAdc) | $V_F$ | –<br>–<br>–<br>– | 900<br>1000<br>1100<br>1250 | mV |
| Diode Capacitance ($V_R$ = 0 V, f = 1.0 MHz) | $C_D$ | – | 2.0 | pF |
| Reverse Recovery Time ($I_F$ = $I_R$ = 10 mAdc) (Figure 1) | $t_{rr}$ | – | 3.0 | μs |




1. A 2.0 kΩ variable resistor adjusted for a Forward Current ($I_F$) of 10 mA.
2. Input pulse is adjusted so $I_{R(peak)}$ is equal to 10 mA.
3. $t_p$ » $t_{rr}$

**Figure 1. Recovery Time Equivalent Test Circuit**

## TYPICAL CHARACTERISTICS

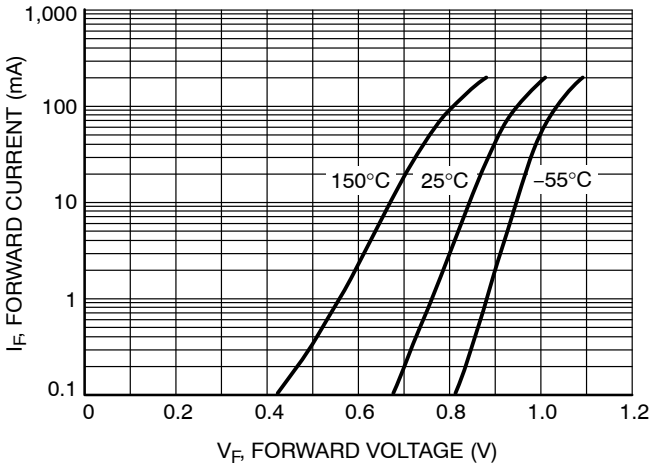



**Figure 2. Forward Voltage**

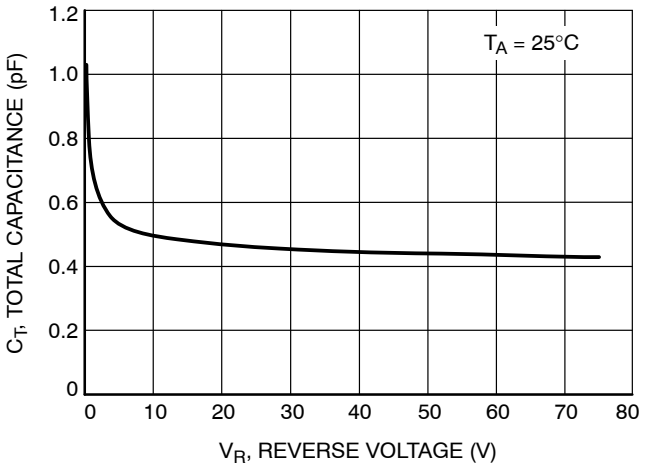
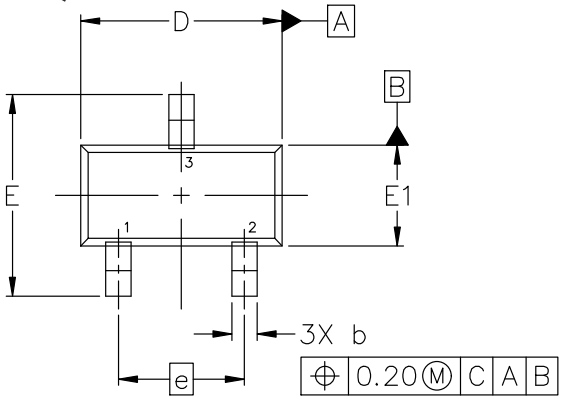



**Figure 3. Capacitance**

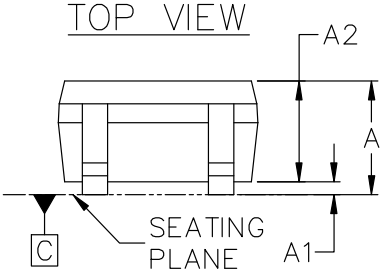
**SC75−3 1.60x0.80x0.80, 1.00P**
CASE 463
ISSUE H

DATE 01 FEB 2024

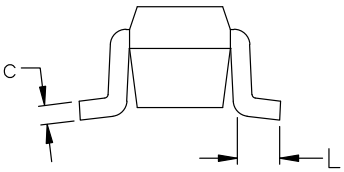





TOP VIEW




SIDE VIEW



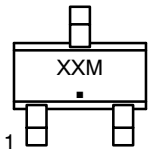

END VIEW

NOTES:

1. DIMENSIONING AND TOLERANCING CONFORM TO ASME Y14.5−2018.
2. ALL DIMENSION ARE IN MILLIMETERS.

| DIM | MILLIMETERS | | |
|---|---|---|---|
| | MIN. | NOM. | MAX. |
| A | 0.70 | 0.80 | 0.90 |
| A1 | 0.00 | 0.05 | 0.10 |
| A2 | | 0.80 REF. | |
| b | 0.15 | 0.20 | 0.30 |
| c | 0.10 | 0.15 | 0.25 |
| D | 1.55 | 1.60 | 1.65 |
| E | 1.50 | 1.60 | 1.70 |
| E1 | 0.70 | 0.80 | 0.90 |
| e | | 1.00 BSC | |
| L | 0.10 | 0.15 | 0.20 |

**GENERIC**
**MARKING DIAGRAM***




XX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

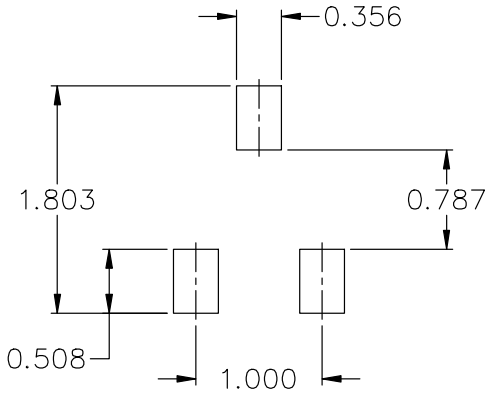
STYLE 1:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 2:
PIN 1. ANODE
2. N/C
3. CATHODE

STYLE 3:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 4:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 5:
PIN 1. GATE
2. SOURCE
3. DRAIN




RECOMMENDED MOUNTING FOOTPRINT*

* FOR ADDITIONAL INFORMATION ON OUR Pb−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

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|---|---|---|
| DESCRIPTION: | SC75−3 1.60x0.80x0.80, 1.00P | PAGE 1 OF 1 |

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